NEXUS
HIGH SCHOOL INNOVATION CHALLENGE

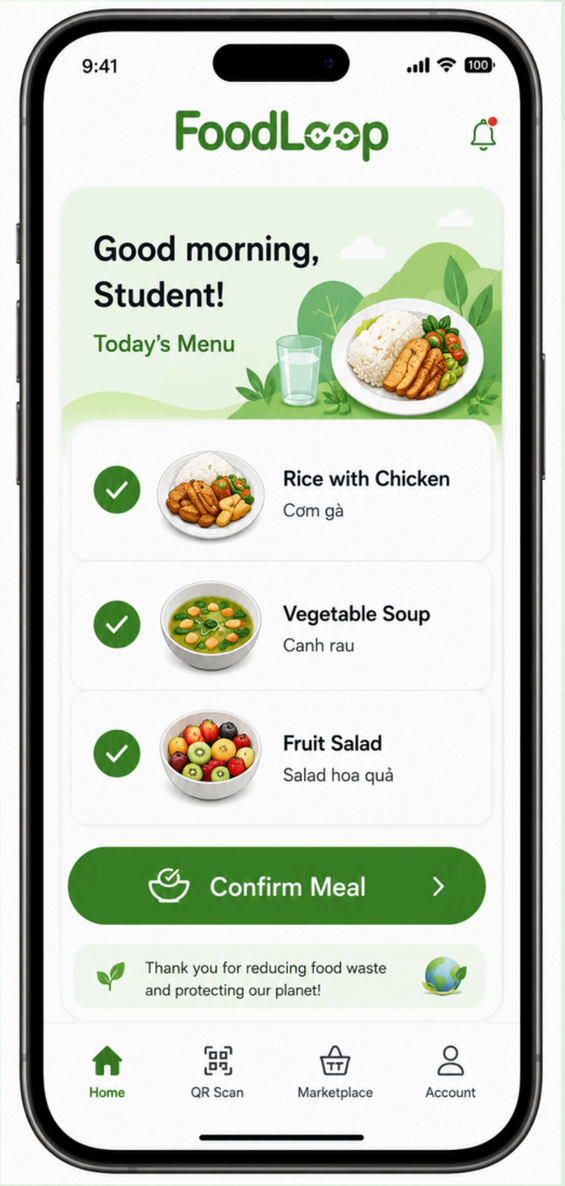
FOOD LOOP

A Solution for Food Waste in High School Canteens

Team Suxen

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Reducing waste. Empowering students. Data-driven canteens.



Problem Statement & Target Users

PROBLEM STATEMENT

1/3 of all food produced globally is wasted annually (FAO). School canteens contribute significantly due to lack of demand-forecasting systems.

Key Causes:

- Students absent from school
- Forgetting to cancel meal reservations
- Intentionally not collecting meals
- No real-time tracking of meal demand

Consequences:

- Increased food waste
- Difficulty in meal planning
- Higher ingredient & labor costs
- Limited data for optimization

TARGET USERS

Students

- Register/cancel meals
- QR code confirmation
- Claim surplus meals
- Earn reward points

Kitchen Staff

- Monitor meal quantities
- Real-time collection tracking
- Reduce food waste
- Optimize ingredients

School Administration

- Monitor reports & statistics
- Manage cafeteria costs
- Evaluate food waste reduction policies









Current Solutions & Limitations

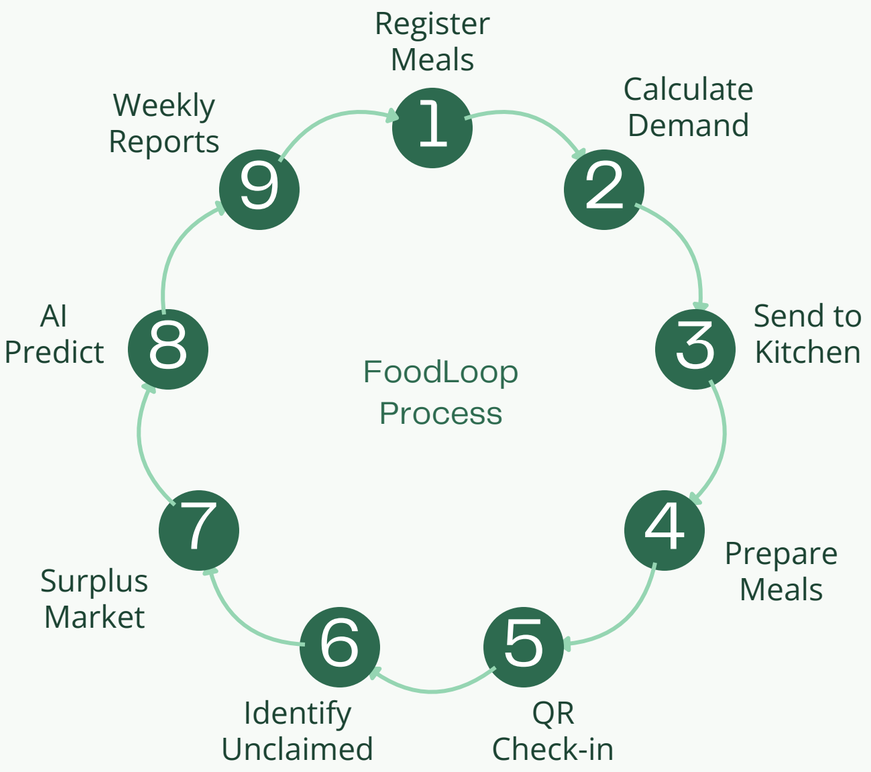
Solution	Limitations		FoodLoop Advantage
Too Good to Go / Olio (Food surplus apps for consumers)	• Food safety issues • No pre-ordering • Not suited for schools	→	• Closed-system for schools • Pre-ordering + QR verification • Safe, controlled environment
Traditional Canteen Software Legacy management systems	• Lacks real-time flexibility • Poor student UX • No AI or surplus marketplace	→	• Real-time updates • Student-friendly mobile app • AI prediction + surplus marketplace
Google Forms / Excel Manual tracking methods	• Slow, time lag • Error-prone, labor-intensive • No analytics or automation	→	• Instant automated updates • Zero manual entry needed • Built-in analytics dashboard

FoodLoop combines real-time meal management, AI prediction, QR verification, and a surplus marketplace in one unified platform.

FoodLoop Solution Overview

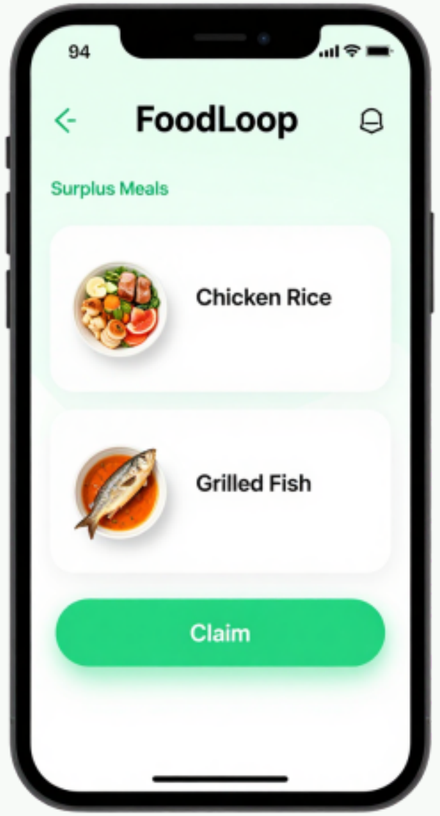
A smart meal management platform connecting students, kitchen staff, and administrators to reduce food waste.

SYSTEM WORKFLOW



KEY BENEFITS

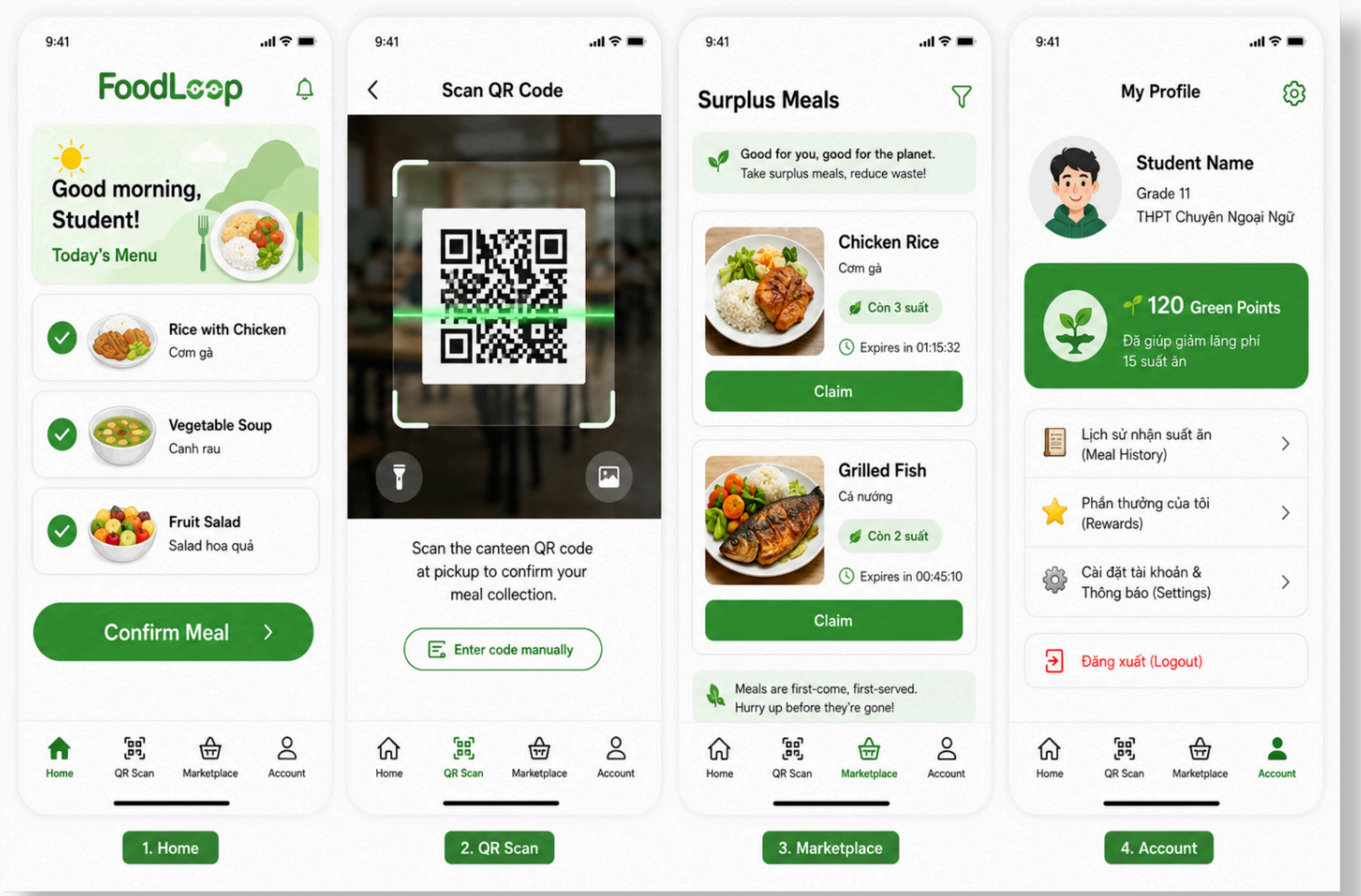
- ✓ Reduces food waste significantly
- ✓ Minimizes unnecessary meal preparation
- ✓ Lowers operational costs
- ✓ Improves cafeteria efficiency
- ✓ Data-driven decision making
- ✓ Encourages student responsibility



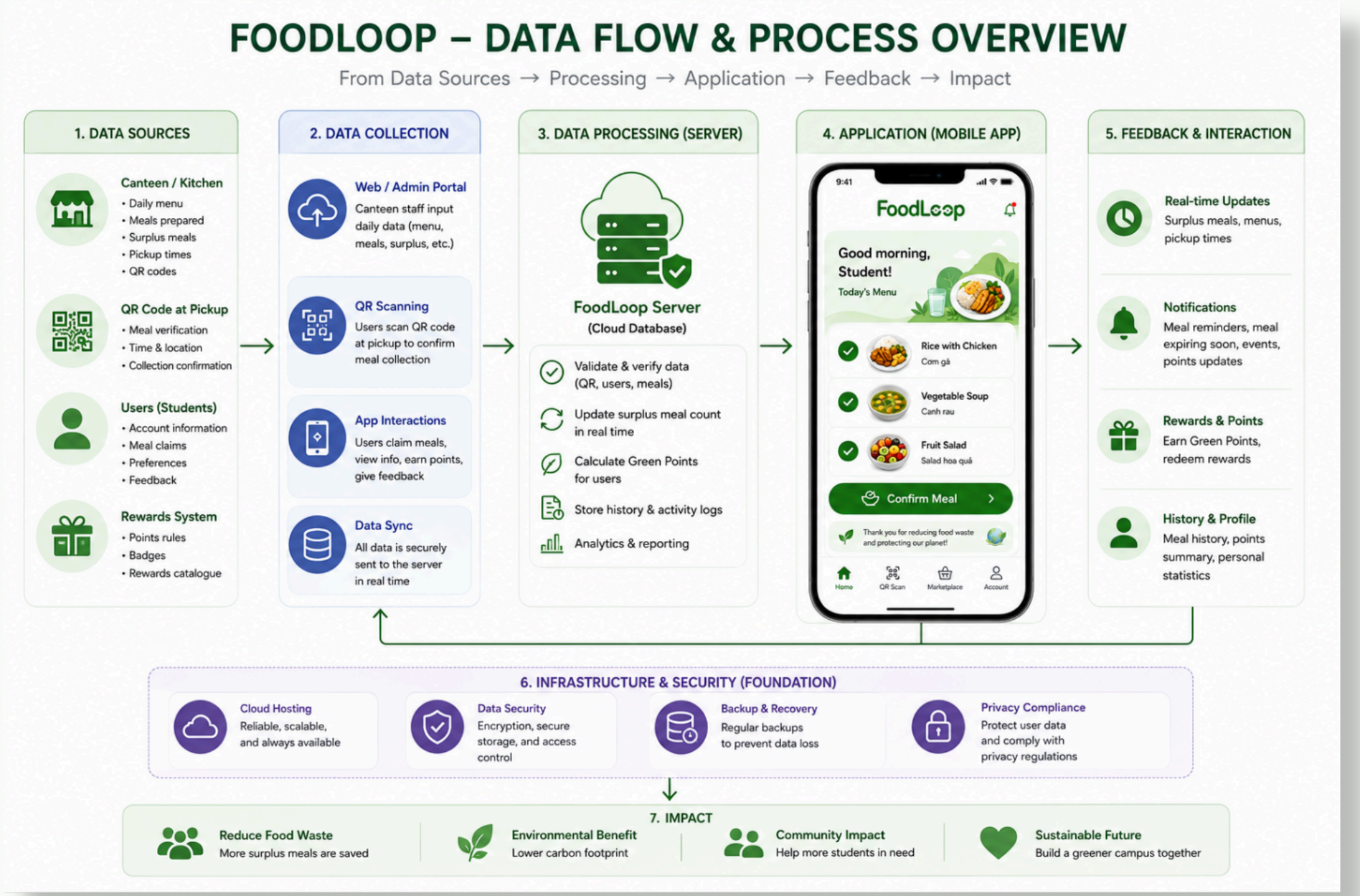
FoodLoop streamlines the entire meal cycle: from registration to AI-powered prediction, ensuring minimal waste and maximum efficiency.

USER INTERFACE OF FOODLOOP

A smart meal management platform connecting students, kitchen staff, and administrators to reduce food waste.



SYSTEM AND DATA WORKFLOW



Key Features of FoodLoop

Core modules that power the FoodLoop ecosystem

1

Flexible Meal Management

- Register or cancel meals before the deadline.
- Students have full control over their daily meal choices.

2

QR Code Check-in

- Scan QR at pickup to confirm collection.
- Automatically filters unclaimed meals for real-time accuracy.

3

Surplus Marketplace

- Unclaimed meals are listed for other students to claim, reducing waste to near zero.

4

AI Meal Prediction

- Analyzes day of week, weather, events, and history to forecast demand accurately.



5

Green Points & Rewards

- Gamification system rewards students who maintain streaks and reduce waste consistently.

6

Dashboard & Analytics

- Weekly/monthly data aggregation helps schools optimize costs and ingredient planning.

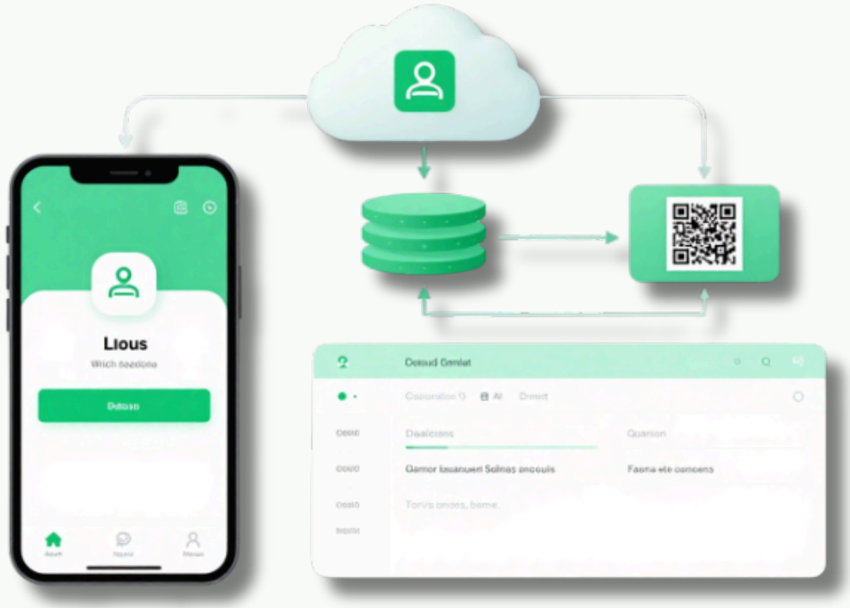
All features are designed to work together seamlessly, creating a closed-loop system that minimizes food waste at every stage.

MVP Scope & Technical Feasibility

What we build first, and the tools we use to build it.

MVP SCOPE

- Use some references from Youtube to know how the school canteen operates
- Pilot with Grade 11 students at one high school canteen
- Core features: Meal registration/cancellation, QR check-in, surplus meal listing
- Basic AI prediction (or manual weekly statistics) for initial testing



TECH STACK

Frontend	Backend	Database	QR System
React.js / Next.js + Tailwind CSS	Node.js + Express.js	Firebase Firestore (real-time)	qrcode.react + html5-qrcode

FEASIBILITY

MVP buildable in 4-6 weeks • 50-100 pilot students • 80-90% technical feasibility • Zero hardware cost

Roadmap & Resources

ROADMAP

Phase 1

Research & Design

2-4 weeks

- Field study at school & kitchen ops review
- UI/UX design
- Finalize MVP feature list

Phase 2

Development & Testing

4-6 weeks

- Code frontend & backend (website prototype)
- Internal testing with 50-100 students
- Pilot kitchen setup

Phase 3

Pilot & Expansion

1-2 months

- Official pilot with one grade level
- Gather feedback & refine AI prediction
- Handover to school administration

RESOURCES

- **Team: 1-2 Developers (Frontend/Backend) + 1 Product/UI-UX Designer (Research + Design)**
- **Tools: Firebase/AWS free tier, existing smartphones for QR scanning**

Expected Impact

Economic Impact

- Reduced food & ingredient costs
- Lower operational expenses
- Better resource management
- Savings redirected to meal quality



Social Impact

- Promotes responsible eating habits
- Student engagement via Green Points
- Creates sustainable school culture
- Collaboration among stakeholders



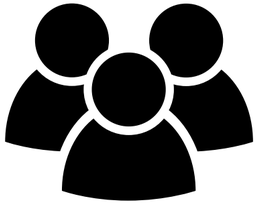
Environmental Impact

- Significantly reduces food waste
- Lowers carbon emissions
- Conserves water & energy
- Sustainable resource use



User Value

- Convenient meal management for students
- Optimized prep for kitchen staff
- Data dashboards for school admin
- Rewards & notifications system



Risk Management & Mitigation

Low User Adoption

Risk: Students forget or choose not to use the app

Mitigation:
Push notifications, Green Points rewards, orientation sessions

Inaccurate AI Predictions

Risk: Insufficient historical data in early stages

Mitigation:
Combine registrations + AI, manual adjustments, continuous learning

Technical Issues

Risk: Server downtime, QR scanning errors

Mitigation:
Regular maintenance, backup servers, manual fallback process

Data Privacy & Security

Risk: Student data exposure

Mitigation:
Encryption, restricted access, minimal data collection, security audits

Implementation Resistance

Risk: Schools reluctant to adopt new systems

Mitigation:
Pilot programs, demonstrate measurable benefits, user-friendly design

Green Points Abuse

Risk: Students gaming the reward system

Mitigation:
QR-based validation only, one point per meal, fraud detection

Bibliography

References and sources used in this project

1. FAO (2011). Global Food Losses and Food Waste. fao.org/4/mb060e/mb060e.pdf

2. ResearchGate (2023). Benefits and Limitations to the Greater Adoption of Food Waste Smartphone Applications.

3. ResearchGate (2024). QR-based Canteen Management System.

Thank You!

FOOD LOOP - Reducing Food Waste in High School Canteens

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